Open, Trusted, and Secure SONiC at Scale



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2016–2023 | OPEN NETWORKING PLATFORM SHIFT

From Azure innovation to industry platform

SONiC's first decade changed the networking operating model: decouple software from proprietary switch hardware, prove it at cloud scale, and turn it into a shared industry platform.

Broke the stack

SAI + containerized NOS separated hardware innovation from software lifecycle.

Proved at scale

Cloud operators could move faster across ASICs, speeds and platform generations.

Created ecosystem

Linux Foundation governance enabled neutral collaboration across clouds, ASICs, OEMs and ISVs.

2016 — **Open source**

Microsoft contributes SONiC + SAI to OCP for open switch platforms.

2018–20 — **Production proof**

Cloud-scale deployments, telemetry, warm reboot and platform breadth mature.

2022 — **LF governance**

SONiC moves to Linux Foundation for neutral community scale-out.

2023 — **Industry platform**

Adoption expands across cloud, enterprise, telecom and software ecosystem.



The breakthrough was not simply "open source."

It was a new network OS model: common software, multiple ASICs, faster hardware adoption, and operations engineered for hyperscale.

SONiC helped shift networking from closed, vertically integrated appliances to a cloud-native, disaggregated platform where the industry can innovate together.

2024–2026 | CLOUD, ENTERPRISE AND AI-SCALE NETWORKING

SONiC enters the AI infrastructure era

The latest chapter is bigger than data-center switching: SONiC is becoming a common operating platform for open, secure, high-performance cloud and AI fabrics.

4,250+
contributors across the project

520+
organizations in the ecosystem

36
Foundation members in 2025



- **Adoption is moving beyond hyperscale** — Case studies span AI supercomputing, telecom, national payments, enterprise and edge use cases.
- **AI is now a core roadmap driver** — SRv6, packet trimming, high-frequency telemetry and lossless/low-latency fabrics address AI workload behavior.
- **Operational quality is becoming industrialized** — Shared labs, regression coverage, workgroups and mentorship turn community scale into production confidence.
- **Security is getting more stringent** — FIPS, Kernel Upgrade, CVE Patches, Secure Boot

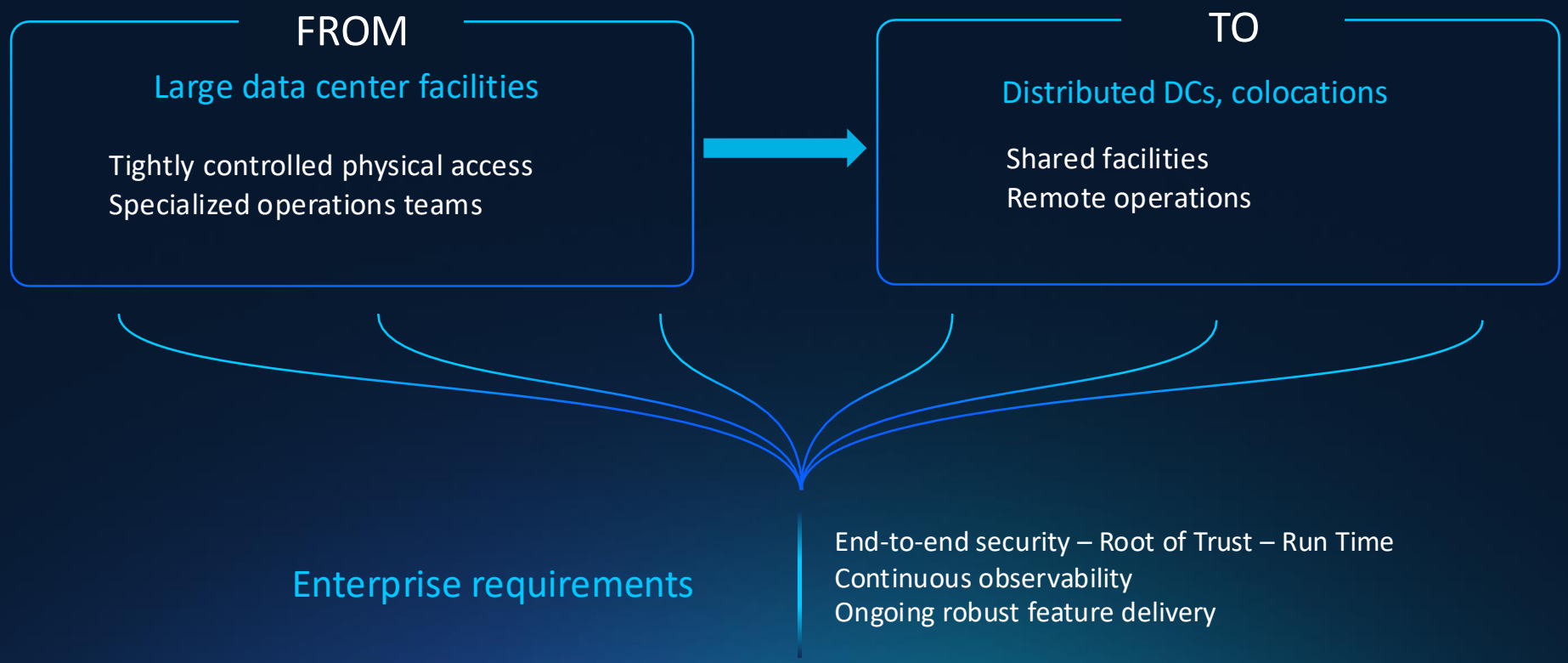
Business impact signals

9 Billion + SONiC enabled switch market in 2028, 20% deployment

40–50%+ TCO/cost savings in 2025 case studies for payments/telecom use cases

Opening close: the next phase of SONiC is about enabling the industry to build faster, more open and more AI-ready infrastructure together.

SONiC is moving into environments with a different risk profile



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Hardware to give SONiC a verified Root of Trust

NOS vendors and hyperscalers can integrate trust primitives without rebuilding the root of trust



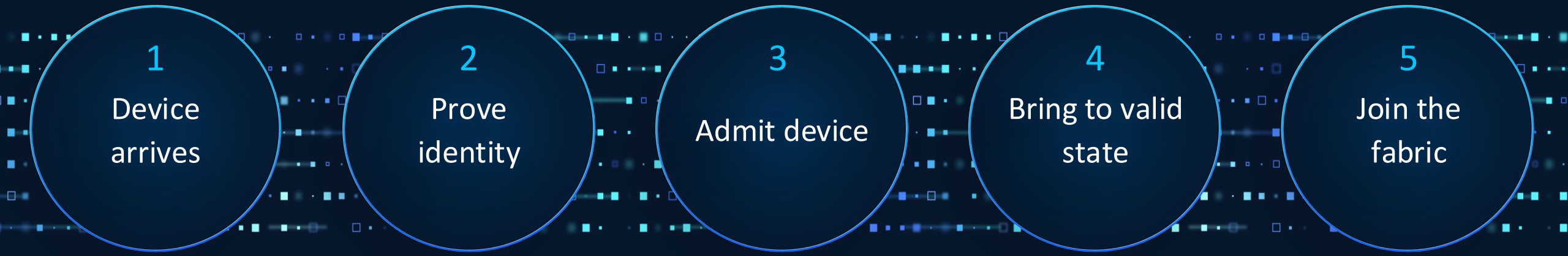
Cisco platform hardware: TAM • Aikido • TPM • OpenTitan • Caliptra

Hardware establishes trust before SONiC even starts

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From assumed trust to policy-driven fabric admission

Identity, attestation and integrity evidence become part of onboarding and operations



Policy options: Allow • Restrict • Quarantine • Investigate

In Production today - Lifecycle management of Cisco Nexus One built upon SONiC

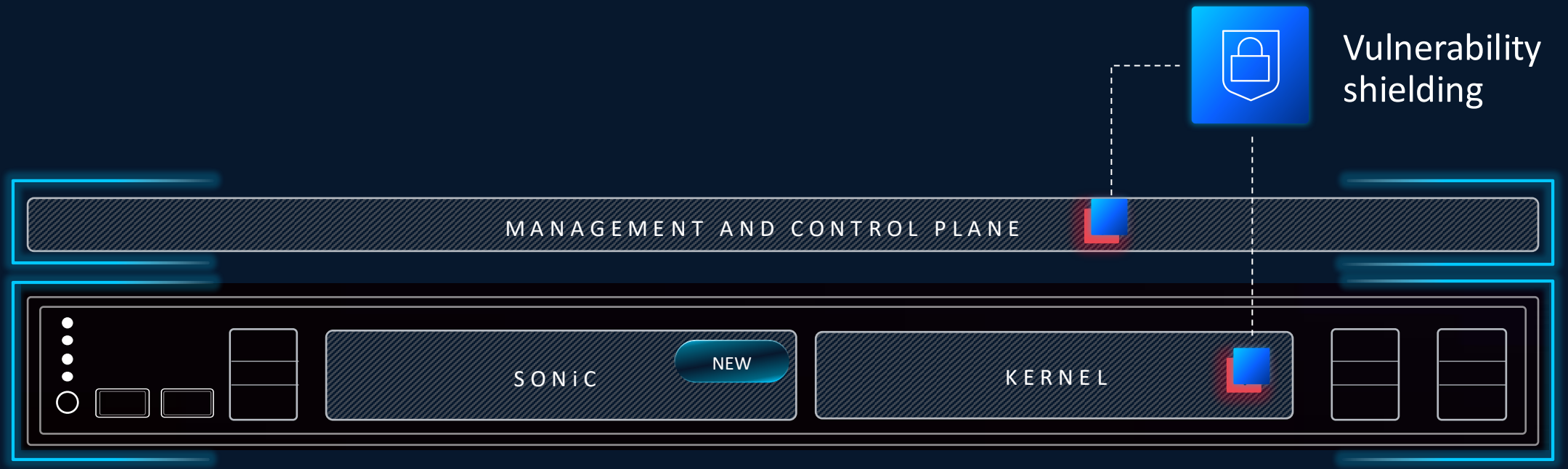
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Stop the attack. Keep the network moving.

Bringing shield to protect networks in production to SONiC

SONiC
Available from Q2’27



Apply policy to live system

Protect with shield now, patch fix in planned window

No downtime or switch reboot

Continuous re-validation

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SONiC software hardening closes the loop in production

Runtime enforcement protects the control plane while the switch is operating in the fabric

ENHANCED

Runtime trust

Detects unauthorized changes to important files while the system runs

Secure containers & packages

Enables to cutomize SONiC while checking images are trusted before they run

Access control

Limits SONiC services to only the actions and resources they need

Secure channels

Encrypts management and telemetry traffic and verifies who is connecting

NEW

Runtime response

Helps block threats and apply security fixes while the system is running

Hardware trust | **SONiC runtime enforcement** | **Operational security**

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Fireside Chat

The Cisco SONiC enterprise security promise

Establish trust. Prove integrity. Recover automatically.

Hardware-rooted trust

Proves the device and its software are trusted, with ownership in the customer's hands

- Secure boot
- Measured boot
- SUDI/IDevID
- Customer ownership

Supply chain assurance

Checks that components and firmware are authentic and have not been altered

- Digital fingerprint
- Signed firmware updates
- Component measurement

Production hardening

Protects software and communications, with security built into development and operations

- Runtime integrity
- Signed containers / packages
- Secure channels
- DevSecOps

Operational resilience

Helps the system recover from failures with backup partitions, boot fallback

- Redundant partitions
- Boot fallback
- Roadmap to multi-disk recovery

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Thank you

